

MOHAMED JRIDI

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Düsseldorf, Germany

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About me: My background is rooted in first principles of both software and hardware. I have great success and expertise in software and embedded development. I am a helpful team player, enthusiastic and cheerful.

Core Focus: Python/Java Ecosystems • SaaS Development • IoT Development • AI Agents

Professional Experience

Freelancer - Düsseldorf

09/2024 - Ongoing

- Strong Understanding of Agentic AI :
 - Building AI Agents, including ChatBots and Browser Extensions (Plugins).
 - MCP Tools, Agents Memory, token optimisation and guardrails.
- Making landing pages for small businesses using Svelte and NextJS.

Lead IoT Developer - Livello GmbH, Düsseldorf

09/2023 - 08/2024

Scaling a cloud-to-edge platform for more markets and building the engineering culture. Fleet Doubled x2.

- Team Mentorship: Leading a team of 5 developers, I focused on code reviews and pair programming. Fostering a culture of clean code and shared technical ownership. I was also responsible for recruitment within the team and mentoring working student on their university projects and thesis.
- Main Architect: Designing a growing software system, written in Python, Java (Vert.x) and Typescript. An event-based containerized Microservices architecture bonded with GraphQL/MQTT/AMQP. A focus on data pipelines, data consistency and reactivity to real-time events.
- Project Management: From Kick-off to delivery, the projects I engage with are systematically tackled using a BDD approach to improve collaboration and communication and meet requirements.
- Projects
 - Refurbished the existing device fleet by porting modern IoT capabilities to legacy hardware, older Raspberry Pi, extending the life span of the product.
 - Integration of RFID smart labels, NFC scanners. Hardware and Java/Python Microservices.
 - Automated OTA for our fridges, using Bash script and Ansible.
 - Modular and automated configuration based provisioning of devices.
 - On device caching with Redis.
- Customer Support: I helped building a new support ticketing system and workflow. I also provided technical assistance and support for client over the phone and on-site.
- Cross-functionality: I am a helpful to all teams assisting them in the issues they face, sharing my expertise and knowledge for them to achieve their projects.

Software Engineer - IoT - Livello GmbH, Düsseldorf

04/2022 - 08/2023

Strengthening code base on our web-platform on GPC and our embedded devices. Automating the most critical parts of our workflow and energizing the company with new talent.

- Code base maintenance: Fixing an old code base and architecture in Python and Java. I have improved the code quality significantly with tangible metrics using SonarQube.
- Complete elimination of technical debt allowing the team to work on new features at ease.
- Projects:
 - Software release and update cycle automation. CI/CD pipelines in GitLab, Ansible for devices.
 - Integration of new POS with ZVT protocol.
 - Engineered a fail-over connectivity system using LTE from the ground up.
 - Deployed a Prometheus/Grafana stack for device monitoring.

Release and update automation with CI/CD pipelines in GitLab and Ansible.

Software Developer in IoT and Backend.

- Maintaining software code.
- Resolving bugs.
- Critical processing speed optimization that significantly impacted the product.

As a Full-Stack developer, I developed a Spring Boot/Angular web application for B2B car rental business.

Education

Master's Degree - Engineering studies, Engineering Degree in Software Engineering, ESPRIT School of Engineering

- Exchange Student at Fachhochschule Schmalkalden, Thuringen - 2020
 - Program: 6 Months
 - Faculty of Computer Science
 - Average Score 1,48 : Very good

Bachelor's Degree in Fundamental Computer Science, FSB Tunisia - 2018

Abitur (Baccalauréat) in Computer Science, 16/20 Very Good. 2011-2012

Main Technical Skills

Programming & Languages

Python : FastAPI/Django, Java : Spring Boot/Vert.x, Javascript : Node.js/Angular/React

Databases and Protocols:

MongoDB, Redis, PostgreSQL, AMQP (RabbitMQ), MQTT, REST, GraphQL, BLE, RFID

Cloud & DevOps

Google Cloud Platform (GCP), Docker, Kubernetes, Ansible, ClearBlade, Prometheus, Grafana

Systems & Networking

Linux System Administration, ARM Architecture, OpenVPN, Raspberry Pi OS, RTOS, tcpdump

Development Boards

Raspberry Pi 3/4B/4B+/Pico, Arduino, STM32

Tools

Github, GitLab, Confluence, Jira, Google Workspace

Methods & Architecture

Agile/Scrum, Microservices, Event-Driven Architecture, Reactive Programming, BDD, TDD, UML

Artificial Intelligence

LLM APIs (Gemini, DeepSeek, Mistral), FastMCP, PydanticAI, Context Engineering, RAG, Agent Memory

Languages

Arabic: Mother tongue

German: B1/ Conversational level

English: Fluent/ Professional level

French: Fluent/ Second language

Hobbies

Football/Padel-Tennis

Calisthenics

Cooking